

Riz Lab

Screw gauge: Copper wire

Aim: To measure the diameter of a given copper wire using a screw gauge.

Apparatus: Screw gauge, a piece of copper wire

Least count: Pitch = 0.5 mm, N = 50, LC = pitch / N = 0.5 mm / 50 = 0.01 mm

Zero error: +0.04 mm (n = 4: +4 × 0.01 mm); zero correction = −0.04 mm

Observations: Diameter (d)

S.No	Direction (∥/⊥)	PSR (mm)	CSR (n)	n × LC (mm)	Observed (mm)	Corrected (mm)
1	∥	0.50	26	26 × 0.01 = 0.26	0.76	0.72
2	∥	0.50	26	26 × 0.01 = 0.26	0.76	0.72
3	∥	0.50	26	26 × 0.01 = 0.26	0.76	0.72
4	⊥	0.50	24	24 × 0.01 = 0.24	0.74	0.70
5	⊥	0.50	24	24 × 0.01 = 0.24	0.74	0.70

Calculations: Diameter (d)

1. **Mean value:** $\bar{d} = (d_1 + d_2 + \dots + d_n) / n$

$$\bar{d} = (0.72 + 0.72 + 0.72 + 0.70 + 0.70) / 5 = 0.7120 \text{ mm}$$

2. **Absolute error of each reading:** $|\Delta d_i| = |\bar{d} - d_i|$

$$|\Delta d_1| = 0.0080, |\Delta d_2| = 0.0080, |\Delta d_3| = 0.0080, |\Delta d_4| = 0.0120, |\Delta d_5| = 0.0120 \text{ mm}$$

3. **Mean absolute error:** $\Delta \bar{d} = \Sigma |\Delta d_i| / n$

$$\Delta \bar{d} = (0.0080 + 0.0080 + 0.0080 + 0.0120 + 0.0120) / 5 = 0.0096 \text{ mm}$$

4. **Relative error:** $\delta d = \Delta \bar{d} / \bar{d}$

$$\delta d = 0.0096 / 0.7120 = 0.01348$$

5. **Percentage error:** $\delta d \times 100 \%$

$$1.35 \%$$

6. **Reporting rule:** $\Delta d = \max(\Delta \bar{d}, LC)$, rounded to the LC's decimal place; $\bar{d}$ rounded to the same place

$$\Delta \bar{d} = 0.0096 \text{ is smaller than } LC = 0.01, \text{ so the uncertainty is the least count: } \Delta d = 0.01 \text{ mm}$$

7. **Result:** $d = (\bar{d} \pm \Delta d) \text{ mm}$

$$d = (0.71 \pm 0.01) \text{ mm}$$

Cross-sectional area of wire

1. **Formula:** $A = (\pi/4)d^2$

$$A = (\pi/4) \times (0.7100)^2$$

2. **Value:** A

$$A = 0.3959 \text{ mm}^2$$

3. **Relative error (maximum-error rule):** $\Delta A/A = 2 \Delta d/d$

$$\Delta A/A = 2 \times 0.01000/0.7100 = 0.0282$$

4. **Percentage error:** $(\Delta A/A) \times 100 \%$

$$2.82 \%$$

5. **Absolute error:** $\Delta A = (\Delta A/A) \times A$

$$\Delta A = 0.0282 \times 0.3959 = 0.0112 \text{ mm}^2$$

Result

- **Diameter (d):** $d = (0.71 \pm 0.01) \text{ mm}$
- **Cross-sectional area of wire: A = (0.396 ± 0.011) mm²**

Precautions

1. The screw was always turned with the ratchet, so the object was never squashed.
2. The final approach was always made in the same direction, to avoid backlash error.
3. The zero error was noted with the faces just touching (ratchet clicking) and the correction applied with its sign.
4. Readings were taken at several points and, for a wire, in two perpendicular directions at each point.
5. The scales were read with the eye perpendicular to the datum line, to avoid parallax.